

Active Maintenance Report: coppock_2019b

Coppock (2019, Journal of Experimental Political Science): Avoiding
Post-Treatment Bias in Audit Experiments

2026-08-01

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This repository holds the actively maintained replication code for Coppock (2019), together with the reproducibility report that documents what the original archive did and did not do. It

is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi [10.1017/S1049096521000366](https://doi.org/10.1017/S1049096521000366)) to a set of published archives.

Article	10.1017/XPS.2018.9
Replication archive	10.7910/DVN/6NVI9C
Underlying data (White, Nathan and Faller 2015)	10.7910/DVN/28158

The data are not redistributed here. The deposit is 1.3 MB across four files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or set of in-text quantities, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See LICENSE.

To reproduce. Clone or download the repository, open `coppock_2019b.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its four files, and produces the published table and every in-text quantity into `maintained/output/`. Required packages: tidyverse, estimatr, rsample, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about twelve seconds, most of it in the 3,000 bootstrap replicates of the bounds estimator. A successful run overwrites `maintained/output/`, which is committed: **git diff on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

No. The archive's single script stops partway through on a current R installation, and it stops twice, at two independent points, neither of which produces a helpful error message.

The first failure is the bootstrap. `broom::bootstrap()` still exists, contrary to what the March 2026 pass on this paper recorded, but it builds a grouped data frame in the format `dplyr` used before version 0.8.0, and the `dplyr::do()` on the next line rejects it with `Corrupt grouped_df using old (< 0.8.0) format`. The second failure is downstream of that and would be reached by anyone who fixed the first: the script reads `fit_1$coefficients`, `fit_1$ci_lower` and `fit_1$ci_upper` out of a tidied `lm_robust` object, and current `estimatr` names those columns `estimate`, `conf.low` and `conf.high`. That one fails with `non-numeric argument to mathematical function`, which says nothing about the renamed column. A third defect stops the script on its first line for anyone not working on macOS or Windows: the `load()` call asks for `voterIDexp_data_sept2014.Rdata` and the deposited file is `voterIDexp_data_sept2014.RData`.

Both are mechanical, so the archive is Category 2: resolvable without the author. Ten lines in four places make it run to completion, and 18 of the 22 published values it can be checked against come back. The 4 that do not are described below, and none of the four is the archive's fault: two belong to R and two to the article's own prose.

1.2 Does the maintained rewrite reproduce the paper?

Yes, on every quantity the paper's own analysis produces. Every estimate, confidence interval and standard error in Table 2 and in the article's prose reproduces to the precision the article prints, and so do the three response rates on the first page, which the deposited script never computed at all.

Three of the 25 checkable rows carry `match_rewrite = 0`, and none of the three is a reproduction failure.

Two are the endpoints of the bootstrap confidence interval on the bounds. R 3.6.0 changed how `sample()` converts uniform draws into integers, so the archive's `set.seed(343)` selects different resamples than it did in 2018. Restoring the old sampler with `RNGkind(sample.kind = "Rounding")` returns the published interval exactly. The March 2026 pass attributed this drift to a difference between `broom::bootstrap()` and `rsample::bootstraps()`; that explanation is wrong, and the two functions are shown below to draw identically.

The third is an inconsistency inside the article. Table 2 reports the Always-Responder estimate as -0.056 and the text on the same page reports it as -5.5 points. The estimate is -5.563 points, so the table rounds and the text truncates. The same truncation appears one sentence later, where the standard error on the redefined outcome is stated as 1.7 points against an actual 1.756.

2 Paper overview

Citation: Coppock, A. (2019). “Avoiding post-treatment bias in audit experiments.” *Journal of Experimental Political Science*, 6(1), 1-4. DOI: 10.1017/XPS.2018.9

Summary: Audit experiments randomize a requester characteristic, such as an apparently Latino versus an apparently non-Latino White name, and measure whether a request draws a response. Researchers routinely want to go further and ask whether the responses that arrive are better or worse, but response is itself a consequence of treatment, so conditioning on it de-randomizes the experiment. The note takes White, Nathan and Faller’s (2015) audit of local election officials as a worked example and lays out the three options an analyst actually has: bound the effect among subjects who would respond either way, find a subgroup that can be assumed to respond either way, or redefine the outcome so that a non-response counts as zero. Applied to the same data, the naive conditional estimate is -3.5 points and not significant, while the redefined-outcome estimate is -6.1 points and the matched-pair Always-Responder estimate is -5.6 points, both significant. Conditioning on response does not merely bias the estimate in principle; here it attenuates it enough to change the conclusion.

3 Original archive reproducibility

The deposit is four files: the analysis script, the White et al. data it reanalyzes, a README, and an HTML log of a 2018 run. There is one script and it produces one table.

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

File	Status on current R	Resolution
aec_replication_JEPS.R	Fails at two points on current R	Replace the <code>do()</code> over <code>broom::bootstrap()</code> ; rename three <code>tidy()</code> columns
voterID-exp_data_sept2014.RData	Loads clean	Correct the case of the filename in the <code>load()</code> call
aec_replication_JEPS.html	Log of a 2018 run; carries Table 2 but no standard errors	None needed
README.txt	Describes the files and the constructed variables	None needed

Every package the script names still installs from CRAN: `dplyr`, `estimatr`, `xtable` and `broom`. Nothing here was retired. All three defects are the same kind of thing, a name or a data structure that moved underneath code that was correct when it was written.

3.1 The bootstrap

The script draws its bootstrap replicates like this:

```
set.seed(343)
boot_df <-
  bootstrap(wnf_subset, 1000) %>%
  do(bounds_wrapper(.)) %>%
  ungroup()
```

`broom::bootstrap()` was deprecated in broom 0.5.0 and is still present in broom 1.0.13, where it warns and then returns a data frame carrying the `indices`, `labels` and `vars` attributes that dplyr used before version 0.8.0. Modern dplyr wants a `groups` tibble instead, so `do()` refuses the object it is handed. The function is not missing and the error does not mention it; the visible failure is two calls downstream of the cause.

3.2 The renamed tidy columns

Once the bootstrap is repaired the script fails again, at the table assembly:

```
estimates <- c(round(fit_1$coefficients[2], 3), ...)
cis <- c(paste0("[", round(fit_1$ci_lower[2], 3), ", ", "...))
```

`fit_1` is `tidy(lm_robust(...))`. Current `estimatr` returns that tidy frame with columns `estimate`, `conf.low` and `conf.high`, following broom's conventions; the names the script uses were the ones `estimatr` returned in 2018. Indexing a data frame by a column that does not exist returns `NULL`, and `round(NULL, 3)` is the error the user actually sees.

3.3 The filename

`load("voterIDexp_data_sept2014.Rdata")` asks for a file the deposit spells `voterIDexp_data_sept2014.RData`. It resolves on macOS and Windows, whose filesystems are case-insensitive by default, and fails on the first line for anyone running on Linux. This defect is invisible to the author and to any reviewer working on the same platform, which is exactly what makes it worth recording.

4 Number-by-number comparison

Table 2: Ground truth: published value against the value the repaired archive script produces on current R. Published values are read from the article; a blank Archive column means the deposited script never computes the quantity.

Location	Claim	Paper	Archive	Match
table_2	naive_estimate	-0.035	-0.035	1
table_2	naive_ci_lower	-0.075	-0.075	1
table_2	naive_ci_upper	0.005	0.005	1
table_2	redefined_estimate	-0.061	-0.061	1
table_2	redefined_ci_lower	-0.095	-0.095	1
table_2	redefined_ci_upper	-0.027	-0.027	1
table_2	ar_estimate	-0.056	-0.056	1
table_2	ar_ci_lower	-0.104	-0.104	1
table_2	ar_ci_upper	-0.007	-0.007	1
table_2	bounds_low	-0.658	-0.658	1
table_2	bounds_high	0.645	0.645	1
table_2	bounds_ci_lower	-0.73	-0.739	0
table_2	bounds_ci_upper	0.726	0.725	0
text_p1	response_rate_non_latino	70.5		
text_p1	response_rate_latino	64.8		
text_p1	response_rate_difference	-5.7		
text_p3	bounds_low_points	-66	-66	1
text_p3	bounds_high_points	65	65	1
text_p3	n_ar_pairs	719	719	1
text_p3	ar_estimate_points	-5.5	-5.6	0
text_p3	ar_se_points	2.5	2.5	1
text_p3	redefined_estimate_points	-6	-6.1	1
text_p3	redefined_se_points	1.7	1.8	0
text_p3	naive_estimate_points	-3.5	-3.5	1
text_p3	naive_se_points	2	2	1

Of the 25 recorded claims, 22 can be compared against something the repaired archive produces. 18 match and 4 do not. The remaining 3 are the three response rates the article’s first page attributes to White et al. (2015), which the deposited script never computes.

5 The bootstrap interval and the R 3.6 sampler

The one published quantity the archive no longer returns is the bootstrap confidence interval on the bounds, and the reason is not in the archive.

R 3.6.0, released in April 2019, replaced the “Rounding” method that `sample()` used to convert uniform variates into integers with a “Rejection” method. Every resample the bootstrap draws therefore differs from the one the same seed drew in 2018. `text_bootstrap_sampler.R` computes the interval three ways to isolate the cause.

Table 3: The bounds bootstrap interval, computed three ways, against the published values. Only the pre-3.6 sampler returns the article’s interval.

Method	Lower	Upper
Maintained rewrite: <code>rsample::bootstraps()</code> , R 3.6+ sampler	-0.7386	0.7254
Archive resampling scheme, R 3.6+ sampler	-0.7386	0.7254
Archive resampling scheme, pre-3.6 sampler	-0.7296	0.7258
Published (Table 2)	-0.73	0.726

Two things follow. The first is that `rsample::bootstraps()` and the archive’s own `sample(n, replace = TRUE)` scheme give the identical interval at the identical seed, to the last digit. The March 2026 pass on this paper recorded the divergence as “expected due to RNG difference between `broom::bootstrap()` and `rsample`” and wrote that explanation into both the notes and the report; the two rows above that agree exactly are the same 1,000 resamples reached by two routes, and the explanation was false.

The second is that restoring the old sampler restores the published interval to the digit the article prints, $[-0.73, 0.726]$. The deposit is internally reproducible. What changed was R.

The maintained rewrite keeps the modern sampler, because reproducing a number by pinning a retired random number generator is a way of preserving an artifact rather than a result. The difference is in any case well inside Monte Carlo noise for a 1,000-replicate bootstrap of an interval more than 1.4 wide.

6 Maintained rewrite

The rewrite lives in `maintained/`: a shared `helpers.R`, one cleaning script, one table script, and three scripts for quantities the article states in prose. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used.

Table 4: Maintained rewrite scripts.

Script	Output	Description
<code>helpers.R</code>	(sourced by every script)	Packages and the Zhang and Rubin bounds estimator
<code>clean_voterID.R</code>	<code>wnf_subset.rds</code> , <code>wnf_always_responders.rds</code>	The two analysis frames: week-one non-bounced emails, and matched-pair Always-Responders
<code>table_2_reanalysis.R</code>	<code>table_2_reanalysis.csv</code>	The four estimators of Table 2, at full precision
<code>text_in_text_estimates.R</code>	<code>text_in_text_estimates.csv</code>	The bounds, the pair count and each estimator in percentage points
<code>text_response_rates.R</code>	<code>text_response_rates.csv</code>	The response rates the first page attributes to White et al. (2015)
<code>text_bootstrap_sampler.R</code>	<code>text_bootstrap_sampler.csv</code>	The bounds interval under three resampling routes

6.1 Architecture

`clean_voterID.R` builds both analysis frames and writes them to `output/`, so the sample restrictions are stated once and every downstream script reads the same rows. `table_2_reanalysis.R` writes its estimates at full precision rather than at the three decimals the article prints, and the two in-text scripts read that file rather than recomputing the models. Nothing in this repository rounds a number twice, and no published value is typed into a script.

The bounds estimator is the one place where the rewrite reimplements rather than replaces. It is the Zhang and Rubin (2003) trimming estimator, which the archive adapted from Aronow, Baron and Pinson’s source code; it exists in no package, so `helpers.R` carries it, translated from base-R indexing and two accumulator loops into `arrange`, `filter` and `cumsum`. It is called from both the table script and the sampler script, which is why it lives in `helpers.R` and not in either.

`text_response_rates.R` is an addition rather than a translation. The article’s opening paragraph reports that non-Latino White names drew a response 70.5 percent of the time and Latino names 64.8 percent, a difference of -5.7 points, attributing the figures to White et al. (2015). The deposited script never computes them, so nothing in the deposit shows they can be recovered. They can, and on exactly the analysis frame the rest of the paper uses.

Table 5: The first paragraph’s response rates, recovered from the deposited data. The article states 70.5, 64.8 and -5.7 .

Quantity	Rewrite	N
Response rate, Putatively non-Latino White name	70.5	1597
Response rate, Putatively Latino name	64.8	1598
Difference in response rates, percentage points	-5.7	3195

6.2 Deprecated patterns replaced

Table 6: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
<code>'rm(list = ls())'</code>	(omitted)
<code>'load("../Rdata")'</code>	<code>'load("../RData")'</code> , the name the deposit carries
<code>'broom::bootstrap()'</code> + <code>'dplyr::do()'</code>	<code>'rsample::bootstraps()'</code> + <code>'purrr::map()'</code>
<code>'fit\$coefficients'</code> , <code>'fit\$ci_lower'</code> , <code>'fit\$ci_upper'</code>	<code>'estimate'</code> , <code>'conf.low'</code> , <code>'conf.high'</code>
<code>'within()'</code> blocks over base indexing	<code>'mutate()'</code> paragraphs with <code>'if_else()'</code>
<code>'left_join()'</code> with no <code>'by ='</code>	<code>'left_join(by = "pair_id")'</code>
<code>'data.frame()'</code> inside the estimator	<code>'tibble()'</code>
two accumulator <code>'for'</code> loops	<code>'arrange()'</code> + <code>'cumsum()'</code>
<code>'xtable'</code> + <code>'print.xtable'</code> (commented out)	<code>'write_csv()'</code> to <code>'output/'</code>
magrittr pipe	native pipe

`xtable` is left alone where a rewrite still needs it; here it is not needed, because the archive’s only `xtable` call is commented out and the table it would have written is reproduced as a CSV.

7 Table 2 as reproduced

Table 7: Table 2 of the article, as reproduced by the maintained rewrite. The article prints the same four estimates; its bounds interval is $[-0.73, 0.726]$, for the reason given above.

Estimand	Estimator	Estimate	95% CI
Undefined	Naive difference-in-means	-0.035	[-0.075, 0.005]
ATE on redefined outcome	Difference-in-means	-0.061	[-0.095, -0.027]
ATE among matched pairs in which both members are Always-Responders	Difference-in-means among matched pairs in which both members respond	-0.056	[-0.104, -0.007]
ATE among all Always-Responders	Bounds	[-0.658, 0.645]	[-0.739, 0.725]

Table 8: The quantities the article states in prose, at full precision. The article rounds the bounds to -66 and 65 points and the standard errors to one decimal.

Quantity	Rewrite
Lower bound, percentage points	-65.780
Upper bound, percentage points	64.539
Matched pairs in which both members responded	719.000
Always-Responder ATE, percentage points	-5.563
Always-Responder SE, percentage points	2.482
Redefined-outcome ATE, percentage points	-6.100
Redefined-outcome SE, percentage points	1.756
Naive ATE, percentage points	-3.542
Naive SE, percentage points	2.041

8 Maintained rewrite verification

Table 9: Maintained rewrite verification: published value against rewrite output.

Location	Claim	Paper	Rewrite	Match	Note
table_2	naive_estimate	-0.035	-0.035	1	
table_2	naive_ci_lower	-0.075	-0.075	1	
table_2	naive_ci_upper	0.005	0.005	1	
table_2	redefined_estimate	-0.061	-0.061	1	
table_2	redefined_ci_lower	-0.095	-0.095	1	
table_2	redefined_ci_upper	-0.027	-0.027	1	
table_2	ar_estimate	-0.056	-0.056	1	

Table 9: Maintained rewrite verification: published value against rewrite output. (*continued*)

Location	Claim	Paper	Rewrite	Match	Note
table_2	ar_ci_lower	-0.104	-0.104	1	
table_2	ar_ci_upper	-0.007	-0.007	1	
table_2	bounds_low	-0.658	-0.658	1	
table_2	bounds_high	0.645	0.645	1	
table_2	bounds_ci_lower	-0.73	-0.739	0	R 3.6.0 changed sample(); under RNGkind(sample.kind = 'Rounding') the same seed gives -0.730, which is the published value
table_2	bounds_ci_upper	0.726	0.725	0	R 3.6.0 changed sample(); under RNGkind(sample.kind = 'Rounding') the same seed gives 0.726, which is the published value
text_p1	response_rate_non_latino	70.5	70.5	1	Attributed to White et al. (2015); the deposited script never computes it. The maintained rewrite recovers it from the deposited data on the paper's own analysis frame
text_p1	response_rate_latino	64.8	64.8	1	Attributed to White et al. (2015); the deposited script never computes it
text_p1	response_rate_difference	-5.7	-5.7	1	Attributed to White et al. (2015); the deposited script never computes it
text_p3	bounds_low_points	-66	-66	1	Text states -66 points; -65.78 at full precision
text_p3	bounds_high_points	65	65	1	Text states 65 points; 64.54 at full precision
text_p3	n_ar_pairs	719	719	1	Not printed by the deposited script; 719 distinct pair_id values in the wnf_always_responders object it builds
text_p3	ar_estimate_points	-5.5	-5.6	0	Text states -5.5 points, Table 2 states -0.056; the estimate is -5.563 points, so the text truncates where the table rounds. Internal inconsistency in the article, not a reproduction failure
text_p3	ar_se_points	2.5	2.5	1	2.482 at full precision
text_p3	redefined_estimate_points	-6	-6.1	1	Text states a 6 point decrease; -6.100 at full precision
text_p3	redefined_se_points	1.7	1.8	0	Text states SE = 1.7 points; the standard error is 1.756 points, which rounds to 1.8. The text truncates rather than rounds
text_p3	naive_estimate_points	-3.5	-3.5	1	-3.542 at full precision
text_p3	naive_se_points	2	2	1	2.041 at full precision

21 of the 25 checkable claims match the published article to the precision it prints. The 4 that do not are the two bootstrap endpoints, which the R 3.6 sampler change explains and the old sampler restores, and the Always-Responder estimate in points, where the article's own text and its own table disagree with each other.

9 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-01

Table 11: Package versions used for the run behind this report.

Package	Version
estimatr	1.0.6
rsample	1.3.2
dplyr	1.2.1
tidyr	1.3.2
purrr	1.2.2
readr	2.2.0
here	1.0.2